

Recovering The Indo-European Language Family Tree Without Cognates: A Comparison of Lexicalized and Delexicalized Representations

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Abstract

The study of historical linguistics was greatly spearheaded by early work on Indo-European languages, which, as one of the largest language families, have been the subject of extensive analysis. Using the Universal Dependencies treebanks dataset on 21 Indo-European languages, we test different lexicalized and delexicalized methods within a simple, unified, and replicable pipeline to reconstruct the Indo-European language tree. Our findings show that, despite differences in corpus content, lexicalized methods perform surprisingly well, that parts-of-speech-based analyses achieve comparable performance; and that despite poor performance, even the simplest baseline methods recover core similarities between languages.

1 Introduction

Ever since scholars first remarked on similarities between languages such as Italian and Sanskrit in 1585 (Kapovic 2017), different works have reconstructed the Indo-European language family tree with a relatively high degree of certainty and consensus with respect to the main subgroups (Olander 2022). Although debates remain as to the positioning of certain sub-branches and the timeline and origins of the tree (Fortson 2010), most sources agree on the overall structure, which we consider our "ground truth" (Figure 1) (Gray & Atkinson 2003).

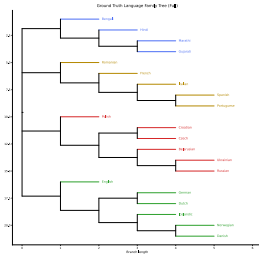


Figure 1: Ground-truth Indo-European phylogenetic tree.

However, the Indo-European language family represents an exception in terms of both available data and agreement on its genealogical structure (Clarkson 2007). Many other language families,

such as Niger-Congo (Dimmendaal & Storch 2016) or Sino-Tibetan (Sagart et al. 2019), are less well established, often due to lack of data or limited prior study. Since most existing methods for language tree reconstruction rely heavily on lexical cognates (Meelen et al. 2022) or features extracted from aligned parallel texts (Schrader & Gultepe 2023), the absence of such resources makes reconstruction significantly more difficult, as is the case for certain Southwest Tibetic languages (Dhakal et al. 2024). In this project, we investigate tree construction by comparing simple lexicalized and delexicalized representations of language. In particular, we ask: **when resources such as labeled cognate sets or aligned parallel corpora are unavailable, do lexicalized or delexicalized feature representations better recover genealogical relationships between languages?** This frames the problem as a comparison between content-based and form-based representations: lexicalized (content) methods operate on word identity and meaning (e.g., embeddings or word distributions), while delexicalized (form) methods abstract away from lexical content and capture structural properties such as syntax. We focus on 21 Indo-European languages, using the well-established "ground truth" tree as our reference and syntactic data from Universal Dependencies treebanks. This provides a controlled setting in which both the data and expected tree are well understood, allowing us to benchmark different feature representations. If certain methods recover known relationships more accurately without relying on cognates or parallel data, similar approaches could be refined and applied to less well-understood language families. Our main hypothesis is that lexicalized representations, particularly those relying on semantic information, will be less effective for reconstructing the language tree, as they are sensitive to differences in the content of the underlying corpora. We expect that simple lexical proxies, such as most-frequent-word and word-length distributions, will perform poorly, and that distributional embeddings will also perform poorly due to noise induced by variation in corpus content. By contrast, we expect that

delexicalized representations based on structural properties - such as part-of-speech and dependency patterns - will reconstruct the linguistic tree with higher accuracy relative to the "ground truth".

2 Related Work

Bouckaert et al. (2012), considered a major reference in the field (Silva & Tehrani 2016), used Bayesian phylogeographic approaches to analyze basic vocabulary data from 103 Indo-European languages, modeling language evolution through the gain and loss of cognate words over time to determine the original homeland and expansion timeline of the Indo-European language family.

Nagata and Whittaker (2013) reconstructed Indo-European language relationships using non-native English texts, introducing a hybrid n-gram language model that incorporates both word tokens and POS tags. By replacing most words with their POS tags while retaining key function words, they reduced lexical variation, allowing the model to focus on structural and grammatical patterns rather than topical vocabulary.

Chen and Gerdes (2017) directly employed the Universal Dependencies (UD) dataset for language classification, asking whether UD treebank data allows recognition of language families based on purely structural syntactic (delexicalized) data. Computing classification on around 50 languages, including Indo-European and Sino-Tibetan, they measured the relative frequency distributions of dependency relations and directional dependency distance, finding that clustering successfully distinguished Indo-European languages and their major subgroups, although errors occurred at more fine-grained levels within subfamilies.

Schrader and Gultepe (2023) proposed a language-agnostic approach to measuring language similarity that avoids reliance on explicit grammatical knowledge. They replaced lexical items with POS tags to reduce semantic content influence, then learned language representations using Doc2Vec embeddings and computed similarity via cosine distance. Their results showed that distances between embeddings aligned with known language families and correlated with geographical proximity.

Phylogenetic tree reconstruction methods, widely used in genetics to determine species separation timelines, have inspired language phylogenetic approaches where languages are modeled as evolutionary systems analogous to DNA (Serva and Petroni, 2008). More recent work

has employed data-driven distance metrics, such as lexical distance, as the basis for constructing phylogenetic trees using clustering methods like UPGMA and Neighbor Joining (NJ), with validity often assessed against known ground-truth trees such as Indo-European (Schrader and Gultepe, 2023).

While state-of-the-art NLP methods enable language-agnostic approaches using word embeddings from models like Word2Vec, systematic comparisons between lexicalized and delexicalized approaches remain limited. We address this gap by evaluating multiple lexicalized and delexicalized representations, using consistent distance measures and tree construction methods, comparing their ability to recover known phylogenetic structure.

3 Methods

3.1 Dataset

All models were trained and evaluated on treebanks from Universal Dependencies (UD) v2.15, a multilingual corpus providing syntactic annotations in a cross-linguistic format. We selected treebanks for 21 Indo-European languages spanning four major subfamilies: Germanic (English, German, Dutch, Danish, Norwegian, Icelandic), Romance (French, Italian, Portuguese, Romanian, Spanish), Slavic (Russian, Ukrainian, Polish, Czech, Croatian, Belarusian), and Indo-Aryan (Hindi, Bengali, Gujarati, Marathi). Nepali was excluded due to the absence of publicly available .conllu data files in the corresponding repository.

3.2 Pipeline



Figure 2: Pipeline of our language phylogeny analysis

We started with the UD dataset. We first derive language-level representations from lexicalized and delexicalized feature sets. Then, we calculate the pairwise distances between each language representation, yielding a distance matrix, which we use to build a phylogenetic tree using both UPGMA and NJ methods. The final trees are compared with the baseline using three different approaches Figure 2 (RF, closest and farthest language pair, and average within-subfamily vs between-subfamily)

3.3 Lexicalized Methods

Lexicalized methods represent each language using its surface word forms, capturing distributional patterns in vocabulary use. We began with simple baseline methods, expecting weak performance but testing whether they could still recover broad subfamily groupings.

Top-50 Most Frequent Word Frequencies: For each language corpus, we computed the frequency of the 50 most frequent words and built a vector for each language. Research has shown that, for Indo-European languages, the more frequently a word is used, the slower it evolves (Pagel et al. 2007) - so perhaps, we thought there would be some overlap in the most frequent words across languages, for example with function words.

Word Length Frequency, Most Frequent Character Trigram Overlap: Here, we calculated, for each language corpus, the frequencies of words of length 1, 2, ..., 9, >10, and clustered the languages using the cosine similarity distance metric. The rationale was that it might capture a weak morphological signal of similarity between languages: closely related languages may exhibit similar word-length distributions due to shared morphological patterns or similar word-formation processes, and this could be reflected in comparable length profiles across languages. Also working for morphological signals, for each language, we extracted the 50 most frequent character-trigrams and computed pairwise Jaccard distance based on set overlap, to capture surface-level morphological similarity between languages.

Word2Vec, Word2Vec + PCA: Word2Vec is a multilingual model used to compute word-level embeddings. We first trained the model and then used it to generate word-level embeddings. Our rationale was that languages naturally share words, and that this would help model to learn cross-linguistic information. For the Word2Vec representation, we apply PCA and retain the components that explain 95% of the variance. This reduces noise and redundancy while preserving the main structure of the embedding space.

Unaligned FastText: FastText addresses the limitations of Word2vec, such as assuming each word is an atomic unit. FastText does this directly through the use of character n-gram representations, where morphological variants of the same root share subword structure, pulling them closer together.

3.4 Delexicalized Methods

Here, we used the full structured treebank data for each language corpus, and implemented different methods based on dependency relations and POS tags.

Frequency-based Methods Building on previous work using frequency distributions to reconstruct language family trees (Chen & Gerdes 2017), we implemented three methods using frequency vectors as feature representations: (1) POS tags, (2) dependency relations, and (3) their concatenation. These features capture very light syntactic structure independent of lexical content.

Transitions Given the proven effectiveness of POS trigrams at representing distances between languages (Gregorio et al. 2024), we computed conditional transition probabilities using two methods: (1) POS tag transitions and (2) dependency relation transitions. For each language corpus, we calculated a feature vector of conditional transition probabilities $P(\text{tag}_2 \mid \text{tag}_1)$ for each POS bigram, and similarly $P(\text{rel}_2 \mid \text{rel}_1)$ for each dependency relation bigram. “

Dependency Distance and Direction Similarly inspired by Chen & Gerdes (2017), who showed partial success deriving language family trees using directional distance to head, we constructed for each language a vector containing, for each dependency relation, the mean signed distance to head and mean absolute distance to head. Distances were computed as the difference between head index and token index (excluding root tokens), yielding a feature vector of size $2 \times |\mathcal{R}|$ per language.

3.5 Tree Reconstruction

Phylogenetic tree reconstruction is NP-hard because the tree space can grow exponentially; hence, heuristic algorithms are used in this field, although they might not result in the best possible tree. However, they will end up at a local optimal point, which might also be the global optimum. In this work, we use two widely adopted distance-based methods: UPGMA and Neighbor-joining (NJ), both used in computational biology and linguistics.

UPGMA is a hierarchical clustering algorithm that iteratively merges the closest clusters based on average pairwise distances. It assumes a constant rate of evolution. In contrast, NJ selects pairs to minimize the total tree length, resulting in a more flexible structure. UPGMA uses simple clustering

based on average distances, while NJ adjusts for differences in language divergence. We use standard implementations from Python libraries (Biopython) to construct trees with both methods and compare their behavior across different representations.

3.6 Distance Metric and Evaluation

We use cosine similarity for all vector-based representations. For the trigram overlap method, we use Jaccard distance, as it directly measures set overlap. We evaluate trees using RF similarity relative to a ground-truth Indo-European tree. The RF distance counts the number of splits, or bipartitions of languages, that differ between two trees. We also analyze closest and farthest language pairs, and compare within-subfamily and between-subfamily distances to assess structural consistency.

4 Results

4.1 RF Distance Results

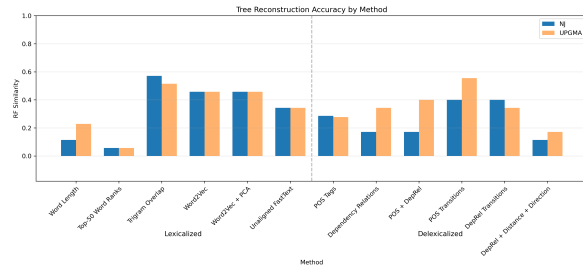


Figure 3: Normalized RF Similarity to Ground Truth by Method

To evaluate how well each inferred phylogenetic tree recovers the ground-truth language family topology, we use the Robinson-Foulds (RF) distance (Robinson and Foulds, 1981). The RF distance between two trees is defined as the number of bipartitions (splits) that appear in one tree but not the other. To facilitate comparison across methods, we report normalized RF similarity.

Figure 3 reports the normalized RF similarity of each method against the ground-truth tree, for both NJ and UPGMA, along with the feature vector dimensionality. The best-performing lexicalized method was top-50 character trigram overlap (Jaccard distance, NJ), and the best-performing delexicalized method was POS tag transitions (NJ). The worst-performing method overall was top-50 word rank frequencies, which produced an essentially arbitrary tree topology.

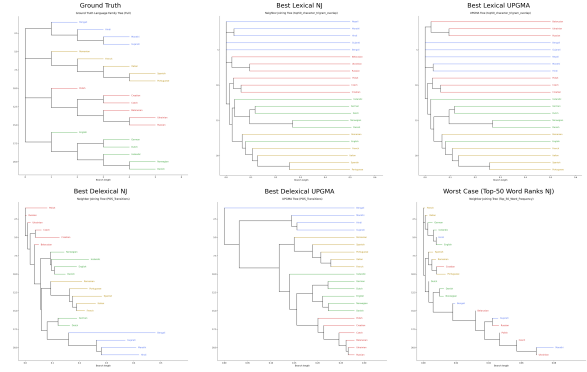


Figure 4: Reconstructed Trees VS Ground-truth

Figure 4 display the six most informative inferred trees alongside the ground truth. Nodes are colored by language subfamily: Germanic, Romance, Slavic, Indo-Iranian.

4.2 General Trends

Lexicalized and delexicalized representations perform at comparable levels. Across both NJ and UPGMA, the best lexicalized methods (trigram overlap, PCA-Word2Vec) and best delexicalized methods (POS transitions) achieved similar RF similarity scores. Unsurprisingly, baseline methods based on word length and top-50 word frequency performed poorly, as they rely only on weak and highly indirect signals. However, even these weakest methods recovered parts of the tree with some accuracy, often systematically grouping certain languages together (e.g., Romance languages). Closest-pair analysis showed lexical methods consistently identify Norwegian–Danish as the closest pair, while structural methods find Russian–Ukrainian or Belarusian–Russian as the closest pair, both linguistically correct.

Feature vector size has limited effect on performance. Methods span dimensionalities from 11 (word length) to 1,600 (dependency transitions), yet larger vectors did not consistently yield better RF similarity. POS tag frequencies (17 dimensions) outperformed dependency relation transitions (1,600 dimensions), indicating that the type of linguistic signal encoded matters more than vector size. Embedding-based methods perform well despite lacking cross-lingual alignment. Word2Vec and unaligned FastText both recovered meaningful family groupings. Across frequency and transition variants, POS-based features produced higher RF similarity than dependency relation features, consistent with literature correlating POS behavior to language family membership.

5 Discussion and Conclusion

The more sophisticated lexicalized methods we chose (Word2Vec, FastText, trigram overlap) performed sufficiently well that they were not negatively affected by the noise introduced by corpus differences, contradicting our initial hypothesis. In fact, the methods that would be expected to be most sensitive to corpus variation (in particular, embeddings) performed among the best, suggesting that lexical and subword-based representations can capture more stable distributional patterns beyond surface semantic variation and retain meaningful genealogical signal even in the absence of aligned data or explicit cognate information.

The character trigram method performed relatively well in grouping languages within subfamilies; however, upon inspection, this was largely driven by script-level similarity. Languages sharing the same writing system (Latin vs. Cyrillic) naturally share character sequences, leading to high trigram overlap. While same-script languages grouped together, the internal structure within these clusters was often incorrect, indicating the method captures orthographic similarity rather than true genealogical relationships.

Word2Vec results must be interpreted with dataset characteristics in mind. Germanic, Romance, and Slavic treebanks are predominantly drawn from news corpora, creating implicit domain alignment that partially compensates for the absence of parallel data. The Indo-Aryan treebanks, by contrast, are smaller and less domain-consistent, which limits the reliability of distributional comparisons for this group. Word2Vec's success is consistent with the distributional hypothesis: languages within the same family share not just vocabulary but distributional context patterns, which are implicitly captured by multilingual co-trained embeddings. (Nagata and Whittaker, 2021)

Among delexicalized methods, POS transitions performed best, consistent with Gregorio et al. (2024). Our interpretation is that transitions capture both how often a tag appears and in what order, giving a richer structural signal than raw frequency alone while remaining low-dimensional enough to be reliable. POS tags likely outperform dependency relations because UD POS tags follow a universal tagset (UPOS), ensuring strict comparability across all 22 languages, whereas dependency relation labels are more granular and may introduce cross-treebank annotation inconsistencies that obscure

typological signal.

NJ and UPGMA results differ as expected. UPGMA assumes a constant evolutionary rate and equal distance from the root, producing more balanced trees, while NJ is more flexible, minimizing overall distance but more prone to noise. In our results, UPGMA performs better on POS transitions, suggesting these features align with the constant-rate assumption, while NJ performs better on trigram overlap.

5.1 Limitations

Firstly, there is data imbalance and coverage. Certain languages, such as Bengali, had much fewer sentences than others (e.g., 200 vs. 10k+), and although most corpora were drawn from news, this was not the case for all. We only considered 21 languages, whereas there are many more in the Indo-European family, meaning that certain subgroups were entirely absent from our analysis. A second limitation is methodological simplicity. Many of the implemented methods captured relatively weak signals, whether syntactic, morphological, or lexical. Many representations rely on aggregated statistics (e.g., mean embeddings, frequency vectors), which may collapse fine-grained contextual or structural information relevant for distinguishing languages. Furthermore, we maintained a simple pipeline with fixed feature vectors rather than more sophisticated modeling procedures, and did not model branch lengths or divergence times, which are often considered in related work to refine tree structure. Finally, the tree construction methods aim to minimize pairwise distances through local decisions, and thus do not guarantee a globally optimal solution; errors or noise in the distance matrix may therefore propagate and affect the overall tree structure.

6 Statement of Contributions

Design: All authors contributed equally to the project design. Each conducted an independent literature review, shared insights, and collaboratively developed the approach. **Implementation:** The pipeline was implemented collaboratively, with team members taking turns developing and refining different components. **Writing:** The writing was divided equally among all authors.

7 Statement on AI Use

In accordance with the course policy on AI use, large language models (specifically ChatGPT and Claude) were used for supplementary tasks such as

debugging, brainstorming, formatting, and proof-
reading, but were not relied upon as the primary
means of completing the assignment, nor for the
core aspects of the project, including the design,
analysis, and interpretation of experiments.

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